SECTION B

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
8.	(a)			moles of $K_2Cr_2O_7$ = 0.017 mol dm^{-3} (1) concentration = $\frac{0.017}{0.250}$ = 0.0680 mol dm^{-3} (1) ecf possible from incorrect M_r	2		1 1			
	(b)	(i)		$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	1	1				
		(ii)		$Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 7H_2O + 3I_2$ do not accept equation with electrons included	1		1			1
		(iii)	I	starch	1	1				1
			II	Moles dichromate = 0.0680 × $\frac{25}{1000}$ = 1.70 × 10^{-3} mol (1) $1Cr_2O_7^{2-} \equiv 3I_2 \equiv 6S_2O_3^{2-}$ so moles thiosulfate = 6 ×1.70 × 10^{-3} = 0.0102 mol (1) concentration = $\frac{0.0102}{\frac{24.30}{1000}}$ = 0.420 mol dm^{-3} (1) ecf possible from incorrect stoichiometry in equation in (b)(ii)	3		1 1	1	1 1	1 1 1